

Nikolaos (Nikos) Triantafyllou

PHYSICIST, PHD CANDIDATE IN COMPUTATIONAL ASTROPHYSICS AND COSMOLOGY

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Research Interests

Large-scale structure, 21cm cosmology, machine learning, Bayesian statistics, model misspecification, bias quantification

First author Publications (full list [here](#))

PUBLISHED/SUBMITTED

TILING I: Field-level Bayesian Reconstruction of Cosmological Initial Conditions during the Epoch of Reionization

N. Triantafyllou, A. Mesinger, S. Murray, S. Gagnon-Hartman

Searching for a Signature of Turnaround in Galaxy Clusters with Convolutional Neural Networks

N. Triantafyllou, G. Korkidis, V. Pavlidou, P. Bonfini

IN PREPARATION

TILING II: Inferring the build-up of Reionization and unseen Galaxies

N. Triantafyllou, A. Mesinger

21cmVAULT: Multi-tracer Inference of Galaxy Properties and an Extensive Database for the First Billion Years of the Universe

N. Triantafyllou, A. Mesinger

Education

Scuola Normale Superiore

PHD IN COMPUTATIONAL ASTROPHYSICS AND COSMOLOGY

• Advisor: Prof. A. Mesinger

Pisa, Italy

Nov. 2023 - Nov. 2027 (Est.)

University of Crete

M.SC. IN ADVANCED PHYSICS (SPECIALIZATION: ASTROPHYSICS) (HIGHEST DISTINCTION)

B.SC. IN PHYSICS (TOP 3% OF DEPARTMENTAL HISTORY)

• Advisor: Prof. V. Pavlidou

Heraklion, Greece

Sept. 2022 - June 2023

Sept. 2018 - Sept 2022

Research Experience

Scuola Normale Superiore

SUPERVISOR: PROF. ANDREI MESINGER

• Description: Research on Large-scale structure and Cosmology for the EoR. Using **bayesian** techniques (Posterior Density Estimation, Neural Ratio Estimation, Diffusion Models) for **field-level, multi-tracer** inference of the **primordial density field** and galaxy properties at $z > 5$.

Pisa, Italy

2023 - Present

University of Crete and IA FORTH

SUPERVISOR: PROF. VASILIKI PAVLIDOU

• Description: Research on Large-scale structure and Cosmology. Used deep (NNs, **CNNs**, GraphNNs) and shallow learning to probe the **turnaround radius** on the plane of the sky based on mass and line-of-sight velocity simulated data at $z < 1$ to constrain cosmological parameters.

Heraklion, Greece

2021 - 2023

International Conferences

Presentations can be found [here](#). For all conferences, see [here](#).

2026 SKA CD/EoR Science Team Meeting

SPEAKER

Belgrade, Serbia

September 2026

International Conference on Machine Learning for Astrophysics (ML4ASTRO3)

SPEAKER

Valetta, Malta

September 2026

A New Era in Astrophysics: preparing for early science with the SKAO

SPEAKER

Görlitz, Germany

June 2025

Simulation-Based Inference for Galaxy Evolution

SPEAKER

Bristol, UK

May 2025

Grants, Fellowships and Awards

Apr. 2026	PI: Compute Time Allocation for AI & Machine Learning project: "Effect of field-level-conditioning on cosmological and astrophysical inferences", LEONARDO HPC/CINECA	95,000 standard hours
Sept. 2025	Compute Time Allocation, E4 Computer Engineering	4-month HPC access
Sept. 2025	First place of poster session, E4 Computer Engineering	Nvidia L4 GPU
Jan. 2025	PI: Compute Time Allocation for AI & Machine Learning project: "Reconstruction the Initial Conditions of the Universe using Diffusion Neural Networks", LEONARDO HPC/CINECA	80,000 standard hours
Oct. 2023	4-year PhD fully-funded scholarship, SNS/PNRR	
2018-2019	"Chrysanthos and Anastasia Karidis" Bequest Scholarship,	

Teaching Experience

Spring 2025	PhD-level course teaching: Structure Formation in the Early Universe, Creation of a hands-on tutorial for cosmological initial conditions	Pisa, Italy
2020 - 2022	Teaching Assistant, 3 labs (Advanced Physics, Optics & Electromagnetism; 6 months each)	Heraklion, Greece

Skills

Programming & Software

Working with: Python, $\text{\LaTeX}$, HPC systems, database creation, parallelization

Familiar with: C, C++, MATLAB, website building

Languages

English (fluent), Italian (conversational, B2), Greek (native)

Academic

Creativity, problem-solving, data science, teaching, report & proposal writing, organizing group meetings/journal clubs, presentation & communication skills.

Academic Service & Broader Impacts

AstroVisio (INAF, Design Group Italia)

Project that aims to develop an innovative virtual reality tool to help researchers visualize astrophysical data

Contribution: Explaining the challenges that astrophysicists face in visualizing data, during the very first and intermediate stages of the project. Provided astrophysical data from simulations. Created a video demo for the possible use cases of Astrovisio for a relevant presentation of the company.

Other Interests

Music (guitar playing, singing, lyric writing), philosophy, art, movies, road trips, swimming, basketball, wine/beer tastings, bars/pubs